DATA SCIENCE REPORT

PRODUCT TAXONOMY INTELLIGENCE: NLP-BASED PRODUCT CLASSIFICATION & EXPLAINABILITY

Character-Level Feature Engineering, Machine Learning & Classification Performance



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly & ML libraries

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Overview & Business Objective

1.1 PROJECT OVERVIEW

E-commerce platforms and product comparison services manage large catalogs containing thousands of product listings from different merchants. These listings often vary in naming conventions, technical specifications, brand references, and product descriptions.

Manually categorizing such products can be time-consuming and inconsistent, particularly when similar products are listed under different names.

The Product Taxonomy Intelligence project applies Natural Language Processing (NLP) and machine learning to automatically classify products into predefined categories using their titles.

The project covers data exploration, text preprocessing, TF-IDF feature engineering, classical machine learning, TextCNN experimentation, model evaluation, error analysis, and explainability.



1.2 PROBLEM STATEMENT

Product titles frequently contain abbreviations, technical specifications, model identifiers, and overlapping terminology, making accurate classification challenging.

The key problem addressed in this project is:

How can machine learning classify products into predefined categories using product titles while maintaining strong performance across different product types?

The project formulates this as a 10-class supervised text classification problem, with *Category Label* as the target variable.

1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Analyze product listings and identify category-specific text patterns.
- Prepare product titles for NLP-based classification.
- Develop and compare word-level and character-level TF-IDF representations.
- Train and optimize multiple machine learning classifiers.
- Evaluate a TextCNN architecture as a deep learning alternative.
- Prevent information leakage through group-aware data splitting and cross-validation.
- Assess classification performance and investigate prediction errors.
- Interpret model decisions using character-level feature contributions.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, NLP-based feature engineering, machine learning classification, model explainability, and business interpretation to identify product naming patterns, evaluate classification performance, understand model decisions, and explore opportunities for automated product categorization.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Category distribution analysis
- Merchant and cluster analysis

02

NLP & Classification

- Text preprocessing
- TF-IDF feature extraction
- Group-aware validation
- Machine learning & TextCNN modeling

03

Model Explainability

- Character feature importance
- Misclassification analysis
- Decision margin analysis

04

Business Interpretation

- Automated product categorization
- Product catalog consistency
- Ambiguous listing identification
- Classification limitations
- Recommendations for improvement